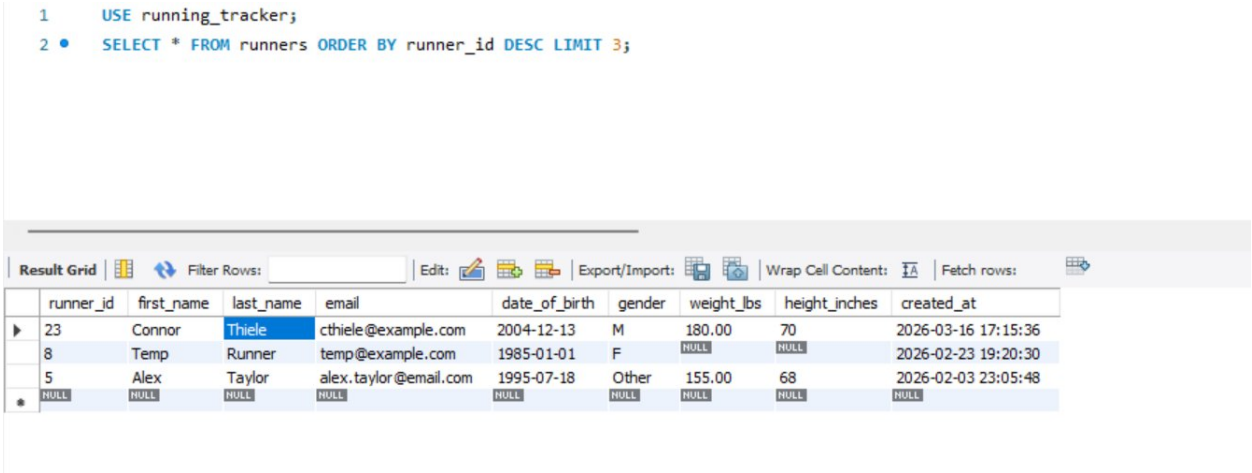
CSCE 548 — Project 4

Full System Test — Runner CRUD Operations

Connor Thiele | github.com/cthiele1/csce548

Test 1: INSERT — New Runner Created in Database

A new runner (Connor Thiele, runner_id = 23) was created using the '+ Add Runner' form in the web frontend. The screenshot below shows the MySQL Workbench query confirming the new record was inserted into the runners table with weight_lbs = 180.00.



```
USE running_tracker;
SELECT * FROM runners ORDER BY runner_id DESC LIMIT 3;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch rows:

runner_id	first_name	last_name	email	date_of_birth	gender	weight_lbs	height_inches	created_at
23	Connor	Thiele	cthiele@example.com	2004-12-13	M	180.00	70	2026-03-16 17:15:36
8	Temp	Runner	temp@example.com	1985-01-01	F	NULL	NULL	2026-02-23 19:20:30
5	Alex	Taylor	alex.taylor@email.com	1995-07-18	Other	155.00	68	2026-02-03 23:05:48
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

*Figure 1 — MySQL Workbench: SELECT * FROM runners ORDER BY runner_id DESC LIMIT 3 showing newly inserted runner #23 (Connor Thiele, weight = 180 lbs).*

Test 2: UPDATE — Runner Record Updated in Database

The Edit button was used to update runner #23's weight from 180 lbs to 190 lbs and height to 90 inches. The MySQL query below confirms the updated values are reflected in the database.

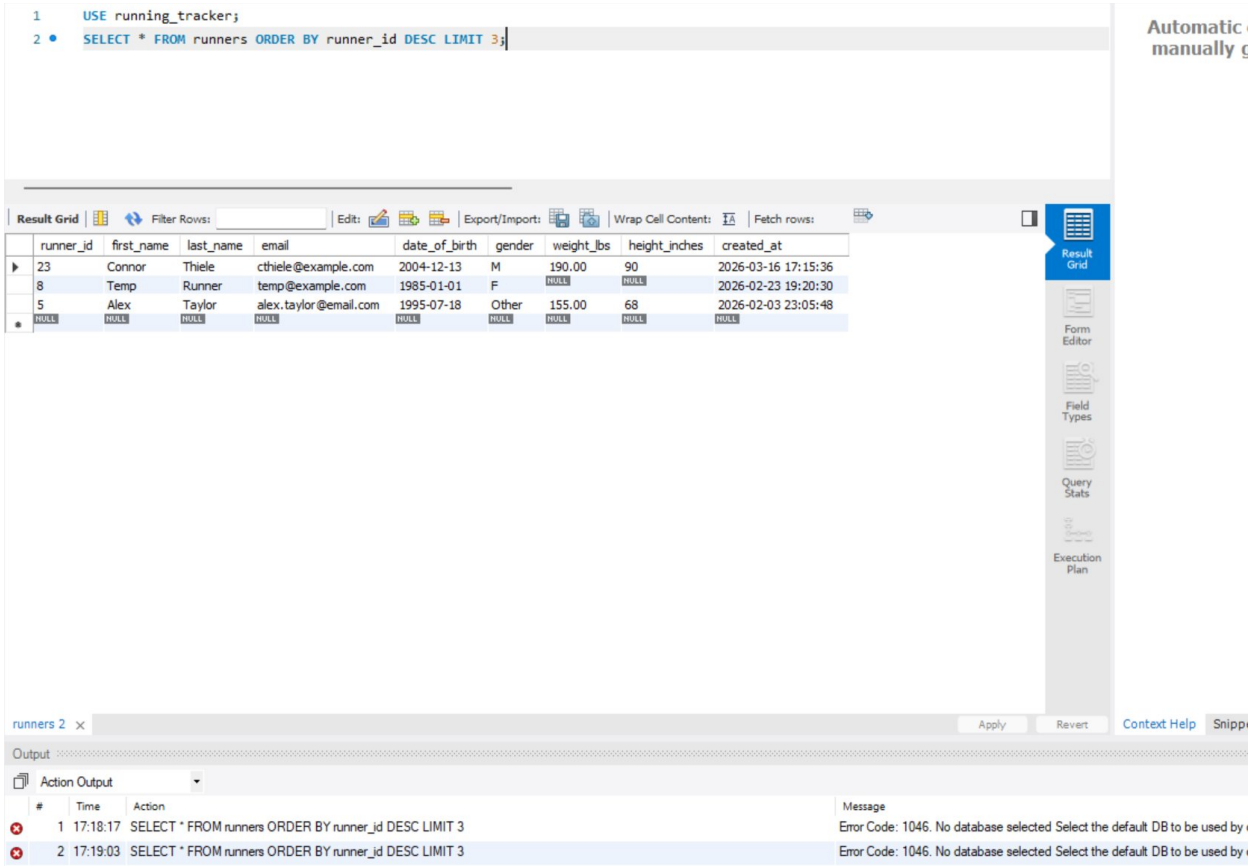
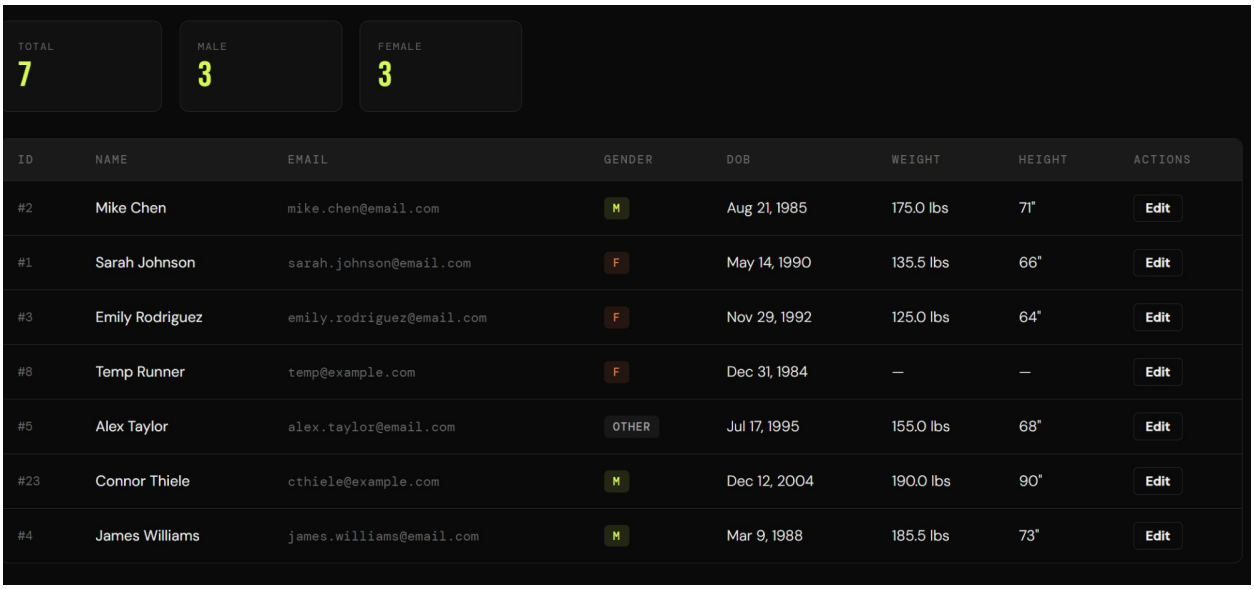


Figure 2 — MySQL Workbench: Runner #23 (Connor Thiele) updated — weight_lbs changed to 190.00, height_inches changed to 90.

Test 3: GET ALL — All Runners Retrieved via Frontend

The Runners page of the web frontend displays all 7 runners retrieved from the database via a GET /runners API call. The updated runner #23 (Connor Thiele) is visible in the table with the updated weight of 190 lbs and height of 90 inches.



TOTAL 7 | MALE 3 | FEMALE 3

ID	NAME	EMAIL	GENDER	DOB	WEIGHT	HEIGHT	ACTIONS
#2	Mike Chen	mike.chen@email.com	M	Aug 21, 1985	175.0 lbs	71"	Edit
#1	Sarah Johnson	sarah.johnson@email.com	F	May 14, 1990	135.5 lbs	66"	Edit
#3	Emily Rodriguez	emily.rodriguez@email.com	F	Nov 29, 1992	125.0 lbs	64"	Edit
#8	Temp Runner	temp@example.com	F	Dec 31, 1984	—	—	Edit
#5	Alex Taylor	alex.taylor@email.com	OTHER	Jul 17, 1995	155.0 lbs	68"	Edit
#23	Connor Thiele	cthiele@example.com	M	Dec 12, 2004	190.0 lbs	90"	Edit
#4	James Williams	james.williams@email.com	M	Mar 9, 1988	185.5 lbs	73"	Edit

Figure 3 — Web Frontend (index.html): All runners table showing 7 records. Runner #23 Connor Thiele visible with updated values. Edit buttons confirm update functionality is available.

Summary

All required operations were successfully demonstrated for the Runners table: INSERT (new runner created via frontend form and confirmed in MySQL), UPDATE (runner weight and height edited via frontend and confirmed in MySQL), and GET ALL (all runners retrieved and displayed in the web frontend). The service layer (Flask API), business layer (validation), and data layer (SQL) all functioned correctly throughout the test.